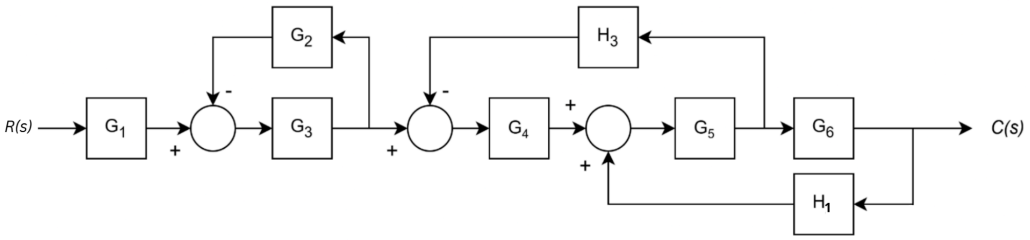
BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 461: Introduction to Robotics

Examination: Quiz III **Duration: 30min**
Semester : Summer 2026 **Full Marks: 15**

Name:	**ID:**	**Section:**

1	 A. **Reduce** the block diagram. Must simplify the final expression.	**7**
2	The automated braking system of a high speed elevator is being tested before an important inspection. The control engineer wants to ensure that the elevator reaches its desired speed smoothly and maintains the target speed with minimal error. For this purpose, a PI controller is implemented and tuned using the Ziegler–Nichols method. At a particular instance, the elevator is moving at 20 m/s, and the desired speed is 35 m/s. During controller testing, when the system was marginal stable, the system was found to have a resulting gain of 0.8 and a critical oscillation period of 10 seconds. a) Determine the appropriate values of **Kp** and **Ki** for the PI controller using the Ziegler-Nichols method. **[2]** b) Using the obtained PI controller parameters, simulate the speed of the elevator over the next **4 steps**. Must show calculation for each step. **[6]**	**8**